

Study Protocol

Automatic-Detection Method for Mining Subsidence Basins Based on InSAR and CNN-AFSA-SVM

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Abstract: Mining subsidence disasters are common geological disasters. Accurate and effective identification of their deformation position is significant in preventing and controlling geological disasters and monitoring illegal mining. In this study, deep learning, combined with a support vector machine (SVM), has been used to establish an automatic-detection method for mining subsidence basins using Sentinel-1A data. The Huainan mining area was selected as the experimental area to verify the method. The interferogram was obtained using differential radar interferometry (D-InSAR) to process the Sentinel-1A radar data of seven landscapes, and the mining subsidence basin and other targets were extracted manually as training samples. Subsequently, AlexNet, VGG19, and ResNet50 convolutional neural networks (CNNs) were used to extract feature vectors of mining subsidence basins for the SVM classifier, and mining subsidence basins were detected in a large-area InSAR interferogram. Non-maximum suppression was used to remove the repeated search box to improve the detection accuracy of mining subsidence basins; the artificial fish swarm algorithm with strong optimization ability and good global convergence is introduced into SVM parameter optimization to construct an improved ResNet50_SVM model. The experimental results show that: (1) the three CNN_SVM methods can accurately detect dry-mining subsidence basins automatically in large regional interference maps, providing an essential scientific basis for the government to monitor illegal mining activities and prevent and control geological disasters in mining areas; (2) the accuracy of the CNN_SVM automatic-detection methods for mining subsidence basins is approximately 80%, and that of ResNet50_SVM for mining subsidence basin detection is 83.7%, superior to that of AlexNet_SVM and VGG19_SVM; (3) the accuracy of the improved ResNet50_SVM based on AFSA algorithm is 88.3%, which is better than the unimproved Resnet50_SVM model.

Keywords: mining subsidence; differential radar interferometry; convolutional neural network; support vector machine



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1. Introduction

The original stress-equilibrium state of overlying strata may be easily broken by mining mineral resources, resulting in strata and surface movement, in addition to deformation. This causes a series of geological and environmental disasters in mines, threatening the safety, life, and property of mining residents [1]. Therefore, to conduct efficient and accurate monitoring of mining activities in mining areas, it is crucial to prevent geological and environmental disasters caused by mining. Moreover, these measures can provide a scientific basis for government departments to supervise illegal mining activities.

Currently, the government mainly monitors illegal mining activities according to the traditional “carpet type” investigation method, and some use microseismic and information

networks. These methods are inefficient and have a small scope for monitoring large-scale illegal mining activities. Despite certain prevention and control measures, illegal mining has been repeatedly banned. According to statistics from the State Administration of Work Safety, China produces 35% of the world's coal mine output, but accounts for 80% of the global death toll caused by coal mining; most mine accidents are caused by illegal underground mining [2]. Therefore, it is necessary to monitor illegal mining activities in mining areas efficiently and accurately [3,4].

Interferometric synthetic aperture radar (InSAR) technology has been widely used in the monitoring of surface subsidence in mining areas, owing to its advantages of low-cost, all-weather, all-day, and high-precision operation [5–8]. Ground subsidence occurs after underground coal mining, and the characteristics of surface deformation are shown in a series of concentric circles or concentric ellipses on the InSAR interferogram, which are called “mining subsidence basins” [9]. Therefore, scholars can monitor illegal mining by detecting mining subsidence basins on the InSAR interferogram. However, with the continuous development of InSAR technology, the image amplitude is also increasing. Visual search alone produces extensive artificial errors and consumes considerable energy in its search for mining subsidence basins in a wide range of interferograms. Therefore, it is necessary to determine a method to find mining subsidence basins in the InSAR interferogram automatically. In recent years, the development of computer hardware and large-scale data collection has helped convolutional neural networks (CNNs) to achieve excellent computer vision results, such as image classification and image detection [10]. Alzubaidi et al. [11] presented a machine-learning-based approach for automatic fracture recognition from unwrapped drill-core images. Zhang et al. [12] proposed a mask-labelling methodology that can establish a large and diverse training set without manual labelling. Xinhong et al. [13] proposed a method, based on the Domain-Adaptive Faster RCNN, called Adaptive-Threshold Cascade Faster RCNN. Liu Fang et al. [14] applied a deep-learning algorithm to the automatic detection and recognition of Oracle bone rubbings to facilitate the research and popularization of traditional culture. CNNs have been widely applied in various industries but have not been applied to mining subsidence basin detection by scholars. Therefore, in this study, we introduce the CNN method for mining subsidence basin detection using a large-width InSAR interferogram.

To effectively monitor illegal mining in large areas and prevent geological disasters, the CNN model, which has achieved excellent results in image detection, is applied to the detection of mining subsidence basins in the InSAR interferogram. By introducing AlexNet, VGG19, and ResNet50 CNN models, and using the support vector machine (SVM) model with strong classification ability to replace the original Softmax classifier of the CNN model, the CNN_SVM automatic-detection method of mining subsidence basin is constructed; the improved ResNet50_AFSA_SVM mining subsidence-basin-monitoring model is constructed by introducing the artificial fish swarm algorithm. This method can effectively detect mining subsidence basins in a large-width InSAR interferogram. In addition, it provides a scientific basis for monitoring mining activities and preventing geological disasters. In addition, it offers essential reference significance for landslides and other geological disasters.

2. Methodology

2.1. CNN Model Principle

A typical CNN model comprises a convolution layer, pooling layer, fully connected layer, and Softmax classification function. A CNN has a vital feature of adaptive extraction, and its mechanism of parameter sharing and inter-layer connection sparsity, introduced within the hidden layer, can significantly reduce the number of model parameters. Three classical CNNs are adopted in this study and their principles are as follows.

AlexNet uses an eight-layer neural network with five convolutional layers, three fully connected layers, and a maximum pooling layer. The entire deep-learning network

contains 630 million links, 60 million parameters, and 650,000 neuron nodes [15]. The AlexNet structural model is shown in Figure 1.

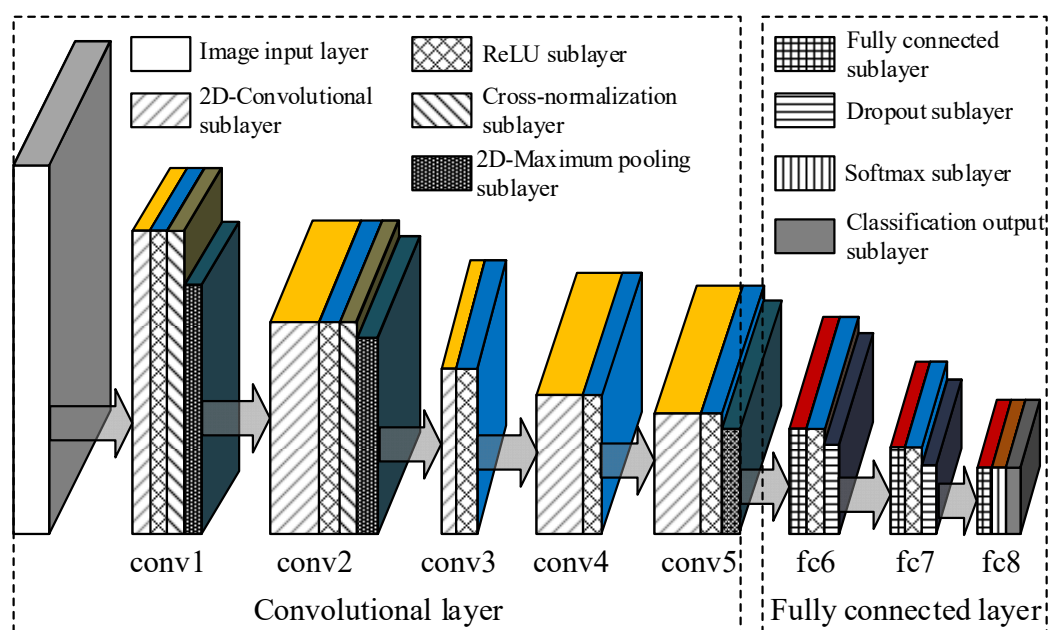


Figure 1. Schematic of AlexNet network structure.

The main difference between the AlexNet architecture and traditional CNNs is the increase in network depth, which leads to an increase in the number of tunable parameters of the model and regularization techniques, such as random inactivation and data enhancement. Random deactivation techniques are applied after the first two fully connected layers in the AlexNet architecture, resulting in less overfitting and better generalization to unknown examples. Another remarkable feature of AlexNet is the use of ReLU nonlinear activation after each convolutional and fully connected layer, which significantly improves the training efficiency compared to the traditionally used hyperbolic tangent function.

VGGNet explores the relationship between the depth of a CNN and its performance; by iteratively stacking small convolutional kernels of 3×3 and maximum pooling layers of 2×2 , VGGNet successfully constructs a CNN with 16–19 layers of depth [16]. The network structure is shown in Figure 2.

VGGNet has five convolutional segments with two–three convolutional layers in each segment; a maximum pooling layer is connected at the end of each segment to reduce the image size. The same number of convolutional kernels is contained within each segment, with more convolutional kernels in the later segments: 64–128–256–512–512. Multiple identical 3×3 convolutional layers are often stacked together, which is a functional design.

The ResNet50 network structure comprises several residual modules. Assuming that x is the input data and with $F(x)$ denoting the residual mapping, the characteristic output, $H(x)$, of the network residual module is:

$$H(x) = F(x) + x \quad (1)$$

When $F(x) = 0$, it means that the convolutional layer performs constant mapping; when $F(x) > 0$, it means that the convolutional layer learns new feature information, ensuring gradient transfer during backpropagation, which effectively solves the problem of gradient disappearance and network degradation during network training [17]. The ResNet50 network structure is illustrated in Figure 3.

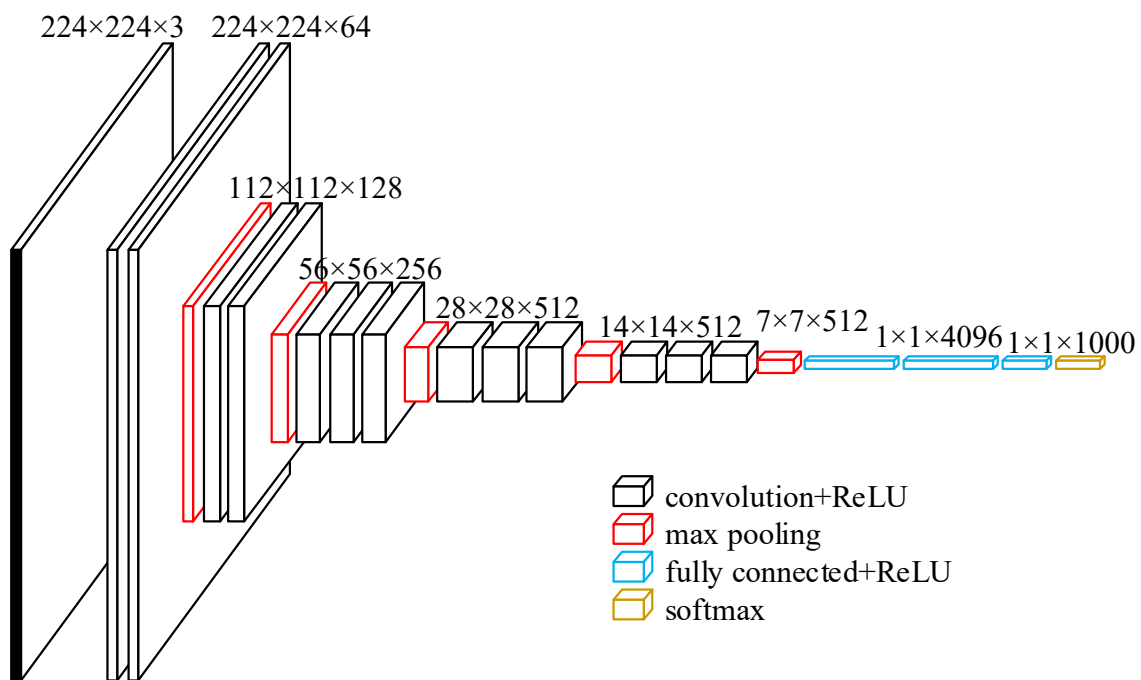


Figure 2. Schematic of VGG19 network structure.

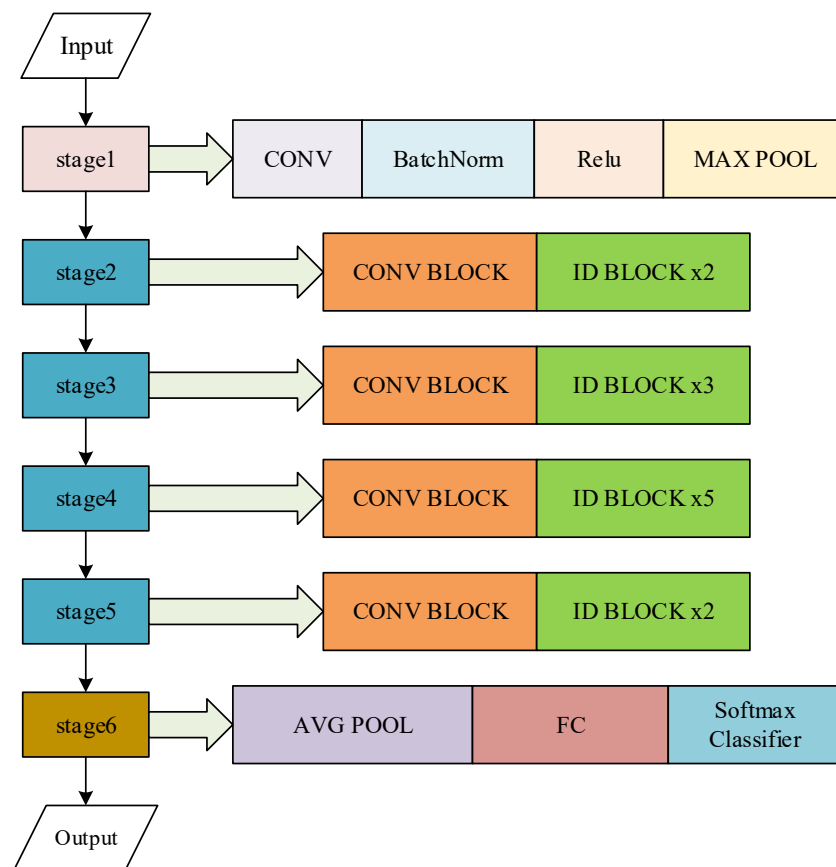


Figure 3. ResNet50 network structure diagram.

The ResNet50 network structure comprises 49 convolutional layers and 1 fully connected layer, and the network operation process comprises six phases. The first stage contains convolution, batch regularization, activation function, and maximum pooling

operations; the CONV (CONVOLUTION) BLOCK, in the second–fifth stages, represents the convolution residual block; the ID (IDENTITY) BLOCK represents the constant residual block; the sixth stage contains the global average pooling layer, the fully connected layer, and the Softmax classifier.

2.2. AFSA Algorithm Principle

The artificial fish swarm algorithm was first proposed by Li Xiaolei et al. in the study of the optimizing mode of animal autonomous bodies [18]. AFSA's rationale is that artificial fish as a whole can be described as $Z = \{X_1, X_2, X_3 \dots, X_i, \dots, X_M\}$, where M is the total number of AF, $X_i = (x_1, x_2, \dots, x_n)$ represents the individual state of AF, and x_i is the variable to be optimized. The food concentration of the current position of the artificial fish is $Y = f(X)$, where Y is the target function value. The distance between artificial fish individuals is denoted as $d = |X_i - X_j|$. *Visual* represents the perceived distance of the artificial fish, *Step* is the maximum stride length of the artificial fish, and δ is the crowding factor, $0 < \delta < 1$.

- (1) Foraging behavior: The current state of the artificial fish is X_i . In its perception range, randomly select a state as X_j , $X_j = X_i + rand(\cdot) \times Visual$, and $rand(\cdot)$ represents any random number between 0 and 1. Compare the food-concentration function, Y , twice. When $Y_i < Y_j$, move one step in this direction; on the contrary, select a state, X_j , for comparison. After repeatedly trying the maximum number of times, if the advance conditions of the artificial fish are still not met, move forward one step at random. The formula is

$$X_{inext} = \begin{cases} X_i + rand(\cdot) \cdot Step \cdot \frac{x_j - x_i}{|x_j - x_i|} & Y_i < Y_j \\ X_i + rand(\cdot) \cdot Step & Y_i > Y_j \end{cases} \quad (2)$$

- (2) Clustering behavior: The number, n_0 , is that of artificial fish in the current field of vision; the position, X_c , is that of artificial fish in the cluster center; and the food concentration, Y_c , is that of artificial fish in the center. When $\frac{Y_c}{n_0} > \delta Y_i$, move one step in this direction, otherwise, conduct foraging behavior, as follows:

$$X_{inext} = \begin{cases} X_i + rand(\cdot) \cdot Step \cdot \frac{X_c - X_i}{|X_c - X_i|} & \frac{Y_c}{n_0} > \delta Y_i \\ \text{Foraging behavior} & \frac{Y_c}{n_0} < \delta Y_i \end{cases} \quad (3)$$

- (3) Rear-end behavior: X_j is the artificial fish with the smallest Y_j in the current field of vision. When $\frac{Y_j}{n_0} > \delta Y_i$, move one step in this direction, otherwise, foraging behavior will be carried out, as follows:

$$X_{inext} = \begin{cases} X_i + rand(\cdot) \cdot Step \cdot \frac{X_j - X_i}{|X_j - X_i|} & \frac{Y_j}{n_0} > \delta Y_i \\ \text{Foraging behavior} & \frac{Y_j}{n_0} < \delta Y_i \end{cases} \quad (4)$$

Firstly, the artificial fish is randomly generated in the parameter interval, the food-concentration function (objective function) is calculated, and the optimal value is recorded. Secondly, the state of each artificial fish after the above three behaviors is compared with the optimal value. If it is better than the optimal value, it will be replaced. After *gen* (total number of iterations) iterations, the state of the artificial fish is the optimal state.

The quality of an SVM algorithm depends on the value of the penalty factor, c , and the kernel-function parameter, g [19]. In this paper, the artificial fish swarm algorithm with strong optimization ability and good global convergence is used to find the optimal penalty factor, c , and the kernel-function parameter, g . The Resnet50_AFSA_SVM mining subsidence-basin-detection model is constructed [20,21].

2.3. Method Construction

First, differential radar interferometry (D-InSAR) is used to process seven scenes of Sentinel-1A radar data to obtain interferograms, and mining subsidence basins and other targets are manually extracted as training samples. Second, AlexNet, VGG19, and Resnet50 convolutional neural networks are used to extract the feature vectors of mining subsidence basins for the SVM classifier. Mining subsidence basins are detected in large area InSAR interferograms, and non-maximum suppression is used to remove repeated search boxes to improve the detection accuracy of mining subsidence basins. The artificial fish swarm algorithm, with strong optimization ability and good global convergence, is introduced into the SVM parameter optimization to build an improved Resnet50_ AFSA_ SVM model. The application process of this method is as follows:

- (1) Constructing sample datasets: Interferograms were obtained by processing Sentinel-1A radar data using differential radar interferometry (D-InSAR), manually cropping the mined subsidence basin as a positive sample dataset and selecting other targets as a negative sample dataset.
- (2) CNN extracts feature vectors: The CNN model is used to extract the features of the mining subsidence basin and other targets, and the extracted feature vectors are input into the SVM classifier.
- (3) SVM classifier: After the feature vector was introduced into the SVM classifier, the artificial fish swarm algorithm searched for the optimal penalty factor, c , and the kernel-function parameter, g , and used the SVM classifier for the training and classification test to test the model accuracy.
- (4) Detection of mining subsidence basin: After the model is trained and tested, it starts to find mining subsidence basins found in the large-width InSAR interferogram, which uses non-maximal value suppression to remove the duplicate search box, and finally outputs the mining subsidence-basin-detection results. The flow of the method is shown in Figure 4.

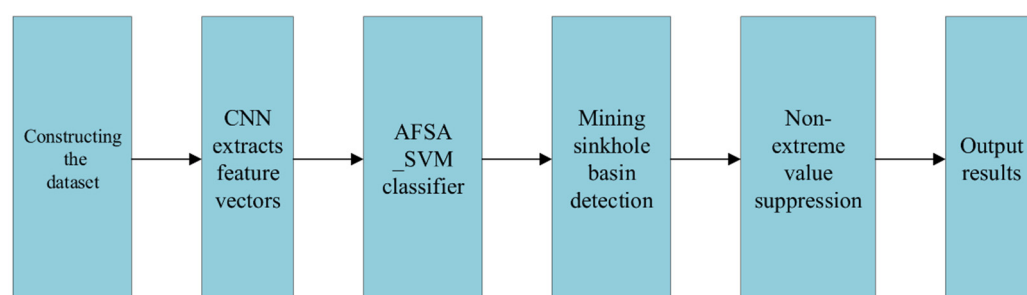


Figure 4. Method overall flow chart.

2.4. Evaluation Criteria

In this study, the precision rate, P , recall rate, R , and $F1$ value are chosen to evaluate the accuracy of the method detection, which is formulated as follows:

$$\begin{cases} P = \frac{TP}{TP+FP} \\ R = \frac{TP}{TP+FN} \\ F1 = \frac{2PR}{P+R} \end{cases} \quad (5)$$

Among them, the meaning of each indicator is shown in Table 1.

Table 1. Meaning of each index.

	Positive Forecast	Negative Forecast
Actual positive	TP	FN
Actual negative	FP	TN

The precision rate represents the proportion of samples classified as positive cases that are actually positive; the recall rate represents the proportion of detected positive samples to the total number of actual positive samples. The value of F1 reflects the comprehensive identification ability of positive and negative samples; the higher the value of F1, the more robust the method [22].

3. Experiments and Results

3.1. Study Area

To verify the method used in this study, the Huainan mining area with many subsidence basins and apparent subsidence was adopted as the test area. The Huainan mining area is located in the north central part of Anhui Province, with a geographical range of approximately $116^{\circ}21'21''$ E– $117^{\circ}11'59''$ E and $32^{\circ}32'45''$ N– $33^{\circ}00'24''$ N, as shown in Figure 5. The Huainan mining area is bounded by the Huaihe River, the Panxie new mining area in the north, and the old mining area in the south. The continuous large-scale mining of the Huainan coalfield has made important contributions to the national economic construction, but has also caused significant ecological and environmental problems [23]. The main structural features of Huainan coalfield are distributed in the NWW direction, while the thrust fault of Huainan coalfield is mainly developed in Huainan. The north and south wings of the syncline are dominated by the south wing thrust fault. Inside the syncline compound, the stratum is flat and open, mainly Carboniferous and Permian, buried under the Cenozoic loose-sedimentary layer; the stratum altitude is gentle. By comparison, the structure in the nappe fault block on the south wing is relatively complex, and the local stratigraphic dip is steep. The overall magmatic rock activity in Huainan Coalfield is not very developed, and the rock mass is relatively less distributed. Yanshanian diorite intrusion is only seen in Panxie mining area.

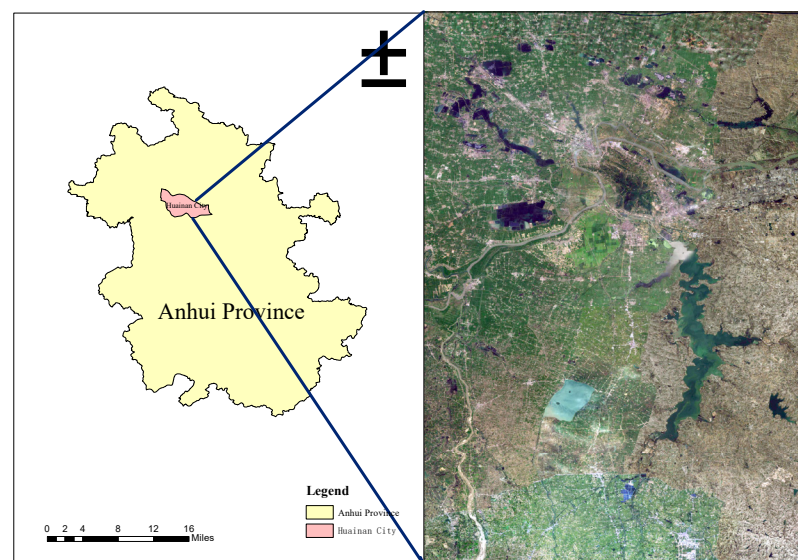


Figure 5. Geographical location of Huainan mining area.

3.2. Built Datasets

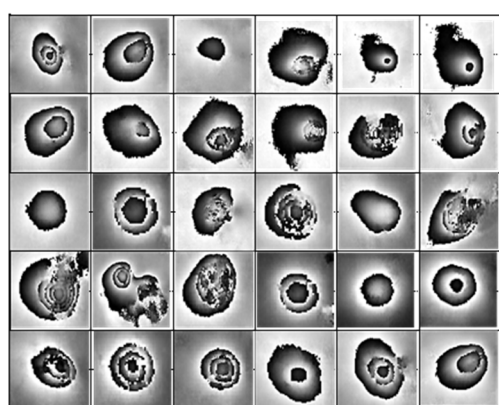
Seven “sentinel-1A” IW-mode images of Huainan mining area, from 16 November 2017 to 27 January 2018, were downloaded in this study and six interference images were generated through DInSAR data processing. The detailed image parameters are listed in Table 2.

From the interferograms formed by the six interference pairs shown in Table 2, the mining subsidence basin targets and other non-mining subsidence basin targets were selected as the sample datasets. A total of 120 mining subsidence basins were considered

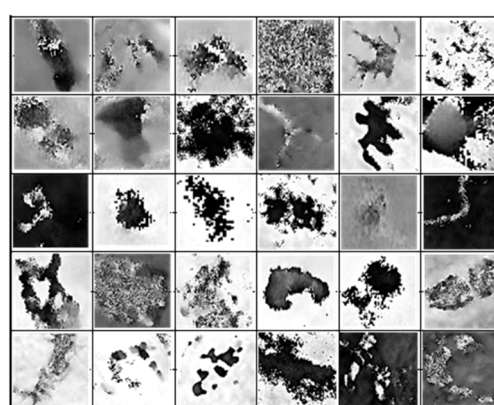
as the positive-sample datasets, and 180 non-mining subsidence basins were considered as the negative-sample datasets. Some sample datasets are shown in Figure 6.

Table 2. Sentinel-1A interference pairs used to construct training samples.

Serial Number	Main Image	Auxiliary Image	Path Number	Time Baseline (D)	Spatial Baseline (M)
1	16 November 2017	28 November 2017	142	12	22.2
2	28 November 2017	10 December 2017	142	12	31.7
3	10 December 2017	22 December 2017	142	12	−44.9
4	22 December 2017	3 January 2018	142	12	−28.6
5	3 January 2018	15 January 2018	142	12	−31.7
6	15 January 2018	27 January 2018	142	12	−50.8



(a) Positive samples



(b) Negative samples

Figure 6. Sample datasets (partial).

3.3. Results and Analysis

In this study, we used 80% of the selected datasets as the training dataset and 20% as the validation dataset. AlexNet, VGG19, and ResNet50 were used for image-feature extraction, and the SVM classifier was used to classify and evaluate the accuracy of the extracted feature vectors. The precision rate, P , recall rate, R , and $F1$ value were selected to evaluate the accuracy of the method detection. The accuracies of the three CNN_SVM models are listed in Table 3.

Table 3. CNN_SVM model accuracy table.

CNN Models	P	R	$F1$
AlexNet_SVM	0.927	0.927	0.927
VGG19_SVM	0.929	0.951	0.940
ResNet50_SVM	0.952	0.976	0.964
ResNet50_AFSA_SVM	0.976	0.976	0.976

From Table 3, we can observe that the accuracy of the above three models, combined with the SVM model for mining subsidence basin detection, is higher than 90%, and the accuracy of the ResNet50_AFSA_SVM model constructed in this paper is up to 97.6%. After the models were trained and tested, large-format wide InSAR interferograms were used for mining subsidence basin detection using the trained models. In this study, interferometric pairs comprising two-view “Sentinel-1A” IW-mode images, from 28 November 2018 to 10 December 2018, were selected for detection, and the results are shown in Table 4. The accuracy rate refers to the ratio of the number of mining subsidence basins detected by the model to the number of all mining subsidence basins in the interferogram; the misdetection

rate refers to the ratio of the number of undetected mining subsidence basins to the number of all mining subsidence basins in the interferogram.

Table 4. Test results.

CNN Models	Number to Be Tested (pc)	Number of Correct Detections (pc)	Accuracy Rate (%)	Missing Detection Rate (%)
AlexNet_SVM	43	33	76.7%	23.3%
VGG19_SVM	43	34	79.1%	20.9%
ResNet50_SVM	43	36	83.7%	16.3%
ResNet50_AFSA_SVM	43	38	88.3%	11.7%

The CNN_SVM automatic-detection method of the mining subsidence basin constructed in this study is applied to mining subsidence basin detection on the InSAR interferogram of the Huainan mining area. The method can detect most mining subsidence basins; the ResNet50_AFSA_SVM model has the highest accuracy of 88.3%. The detection results are shown in Figure 7. In this study, it was found that a small number of undetected mining subsidence basins are mining areas with tiny subsidence areas or are areas with inferior interference quality. The CNN_SVM model constructed in this study is ineffective in detecting when the edge features of mining subsidence basins are not obvious. Scholars further discussed and analyzed this method to improve the detection effect and scalability of the proposed method.

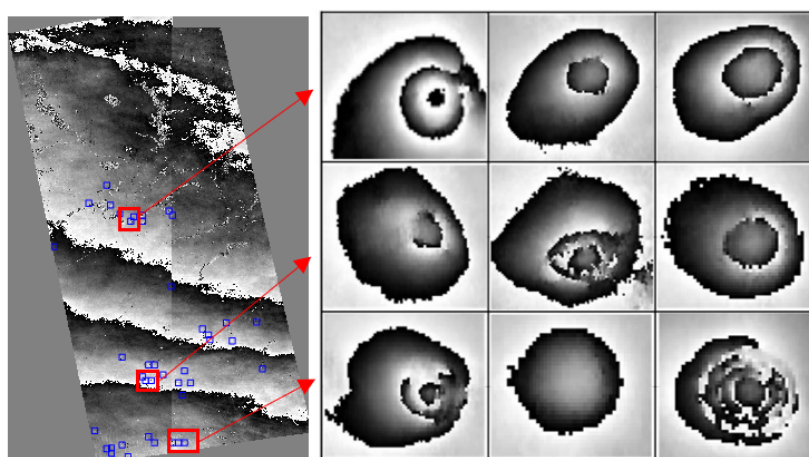


Figure 7. ResNet50_AFSA_SVM model detection result diagram.

4. Discussions

4.1. Impact of Increasing the Datasets in the Method

To discuss the influence of the number of datasets on the CNN_SVM automatic-detection method for mining subsidence basins, owing to the limited number of interference maps of the mining subsidence basin, the dataset of the mining subsidence basin map is expanded by translation and rotation. With 180 mining subsidence basins as positive-sample datasets and 240 non-mining subsidence basins as negative-sample datasets, the experimental results of the ResNet50_AFSA_SVM model are shown in Table 5.

Table 5. ResNet50_AFSA_SVM model expanded dataset detection results.

Datasets	Number to Be Tested (pc)	Number of Correct Detections (pc)	Accuracy Rate (%)	Missing Detection Rate (%)
Original datasets	43	38	88.3%	11.7%
Adding datasets	43	39	90.6%	9.4%

The accuracy of the ResNet50_AFSA_SVM model for detecting mining subsidence basins was 90.6% after expanding datasets, which was higher than 88.3%. It can be observed that the expanded datasets improved the accuracy of model detection, but the detection effect was still poor for some areas with no apparent characteristics of mining subsidence basins.

4.2. Effect of Changing the Proportion of the Datasets on the Method

This study used 80% of the selected datasets as the training dataset and 20% as the validation dataset. To explore whether the change in the ratio of the training datasets and validation datasets is the effect of the CNN_SVM method to detect the mining subsidence basin, in this study, we conducted two experiments for the ResNet50_AFSA_SVM model with +10% of the training datasets and −10% of the validation datasets and −10% of the training datasets and +10% of the validation datasets. The results of the experiments are shown in Table 6.

Table 6. ResNet50_AFSA_SVM model changes the detection result of datasets proportion.

Dataset Proportion	Number to Be Tested (pc)	Number of Correct Detections (pc)	Accuracy Rate (%)	Missing Detection Rate (%)
Original datasets	43	38	88.3%	11.7%
Training datasets +10%	43	39	90.6%	9.4%
Validation datasets −10%	43	36	83.7%	16.3%
Training datasets −10%	43	36	83.7%	16.3%
Validation datasets +10%	43	36	83.7%	16.3%

It can be observed from Table 6 that changing the proportion of datasets has a slight influence on the accuracy of the ResNet50_AFSA_SVM model in detecting mining subsidence basins; the detection effect of mining subsidence basins with obscure image features is still poor.

4.3. Scalability Study of the Method

The method used in this study was successfully applied to the Huainan mining area. Wu et al. [24] proposed the early identification and monitoring of landslides based on InSAR technology and optical remote sensing. Next, scholars will apply the method proposed in this study to detect the surface-deformation characteristics caused by landslides and other geological disasters to monitor and prevent them in the future.

5. Conclusions

- (1) A CNN_SVM automatic-detection method of mining subsidence basins is constructed in this study and can be applied to the detection of mining subsidence basins with large-width InSAR interferograms. The method provides a scientific basis for the government to monitor illegal mining activities and prevent geological disasters in mining areas.
- (2) The method was applied to the Huainan mine area. The experimental results show that the three models detect mining subsidence basins with an accuracy of approximately 80%, and ResNet50_SVM detects mining subsidence basins with an accuracy of 86.0%, which is better than AlexNet_SVM and VGG19_SVM. The accuracy of the improved Resnet50_SVM, based on the AFSA algorithm, is 88.3%, which is better than the unimproved Resnet50_SVM model. Through further discussion and analysis of the method, the accuracy of Resnet50_AFSA_SVM for mining subsidence basin detection is improved by adding datasets. The model constructed in this paper can effectively detect the mining subsidence basin in the large-width InSAR interferogram.
- (3) A small number of mining subsidence basins with inconspicuous edge features are poorly detected by the CNN_SVM mining subsidence automatic-detection method

constructed in this study. The authors will consider increasing the datasets and using higher-resolution InSAR data to enhance the detection effect.

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